

Study of *Hibiscus* Flower**Aim**

To study flower of *Hibiscus* with reference to reproductive structures.

Requirements

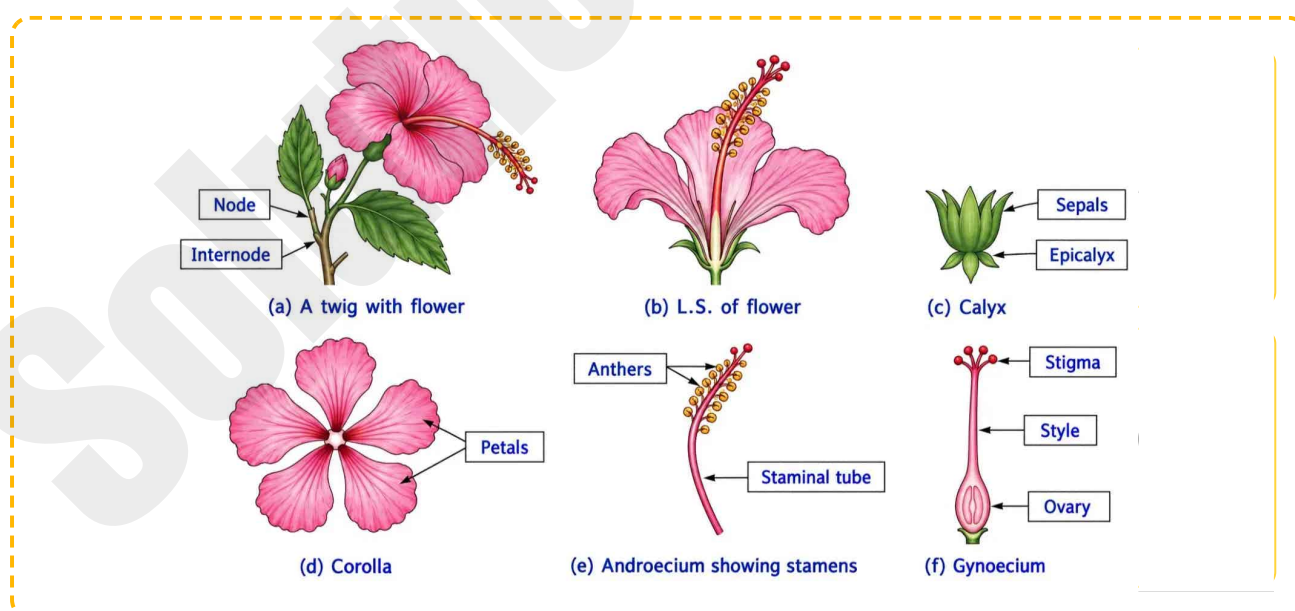
Flowers of *Hibiscus*, forceps, a razor or blade, slides, two needles (pointed and arrow headed), cover slips, a dissecting microscope, a compound microscope, paper, glycerine, etc.

Procedure

1. Take an open flower of *Hibiscus* and observe its different floral whorls, namely calyx, corolla, androecium and gynoecium.
2. Carefully separate out each floral whorl using needles and forceps.
3. Arrange the members of each whorl separately on a white paper or a glass slide.
4. Count the members present in each whorl.
5. Observe the separated members under a dissecting microscope.
6. Carefully separate the stigma from the staminal tube by taking a longitudinal section.
7. Draw diagrams of the members of each whorl.

Diagram

(Draw a neat and labelled diagram of L.S. of *Hibiscus* flower)

**Observation**

Observe different parts of *Hibiscus* flower and enter your observations in the following table:

Whorl of Flower	Number/Member	Description	Functions
Epicalyx	5 to 8 free bracteoles	Green in colour	To protect inner delicate parts in bud condition
Calyx	5 sepals	Green in colour	Protective, keeping all the parts safely covered during bud condition.
Corolla	5, large petals	Red or different coloured, bright and large petals.	To attract insects for pollination.
Androecium	Many stamens	Filaments of stamens are fused to form staminal tube. It bears anthers at its tip.	To produce pollen grains which form male gametes for sexual reproduction.
Gynoecium	5 carpels	Consists of ovary, style and stigma.	To produce ovules which forms female gametes for sexual reproduction.

Inference

What is the function of a flower?

Ans. Flower is the structural unit of sexual reproduction in plants. It attracts insect for pollination.

Multiple Choice Questions

1. The flower of *Hibiscus* is said to be the complete flower because

- (A) it is big in size. (B) it is colourful and thus attracts insects.
(C) it has all the four whorls. (D) it does not develop into fruit.

Answer: (C) it has all the four whorls.

2. Which parts form gynoecium of the flower?

- (A) Ovary, style and stigma (B) Ovary, seeds and fruits
(C) Staminal tube and stigma (D) Calyx covering the ovary

Answer: (A) Ovary, style and stigma

3. How is androecium of the *Hibiscus* flower?

- (A) Free stamens associated with gynoecium
(B) Fused stamens on the petals

(C) Stamens inserted into calyx

(D) Many stamens, filaments of the stamens fused to form staminal tube

Answer: (D) Many stamens, filaments of the stamens fused to form staminal tube

4. Which part of the flower protects it in the bud condition?

(A) Corolla (B) Stamen (C) Bracts (D) Calyx

Answer: (D) Calyx

Remember

Flower is a reproductive organ of a plant. Androecium and gynoecium are the reproductive whorl of a flower which help in formation of fruit and seeds.

Date: _____

Teacher's Signature: _____